Part 1

Linux Device Drivers for Your Girl Friend

This series on Linux device drivers aims to present the usually technical topic in a way that is more interesting to a wider cross-section of readers.

“After a week of hard work, we finally got our driver working,” were Pugs’ first words when he met his girlfriend, Shweta. “Why? What was your driver up to? Was he sick? And what hard work did you do?” asked Shweta. Confused, Pugs responded, “What are you talking about?” Now it was Shweta’s turn to look puzzled, as she replied, “Why ask me? You tell me—which of your drivers are you talking about?”

When understanding dawned on him, Pugs groaned, “Ah c’mon! Not my car drivers—I am talking about a device driver on my computer.” “I know about car and bus drivers, pilots, and even screw-drivers; but what is this ‘device driver’?” queried Shweta, puzzled. That was all it took to launch Pugs into a passionate explanation of device drivers for the newbie—in particular, Linux device drivers, which he had been working on for many years.

Of drivers and buses

A driver drives, manages, controls, directs and monitors the entity under its command. What a bus driver does with a bus, a device driver does with a computer device (any piece of hardware connected to a computer) like a mouse, keyboard, monitor, hard disk, Web-camera, clock, and more.

Further, a ‘pilot’ could be a person or even an automatic system monitored by a person (an auto-pilot system in airliners, for example). Similarly, a specific piece of hardware could be controlled by a piece of software (a device driver), or could be controlled by another hardware device, which in turn could be managed by a software device driver. In the latter case, such a controlling device is commonly called a device controller. This, being a device itself, often also needs a driver, which is commonly referred to as a bus driver. General examples of device controllers include hard disk controllers, display controllers, and audio controllers that in turn manage devices connected to them. More technical examples would be an IDE controller, PCI controller, USB controller, SPI controller, I2C controller, etc. Pictorially, this whole concept can be depicted as in Figure 1.

Device controllers are typically connected to the CPU through their respectively named buses (collection of physical lines)—for example, the PCI bus, the IDE bus, etc. In today’s embedded world, we encounter more micro-controllers than CPUs; these are the CPU plus various device controllers built